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Date: 8-3-2026

Task: Virtualization and Topology Set Up

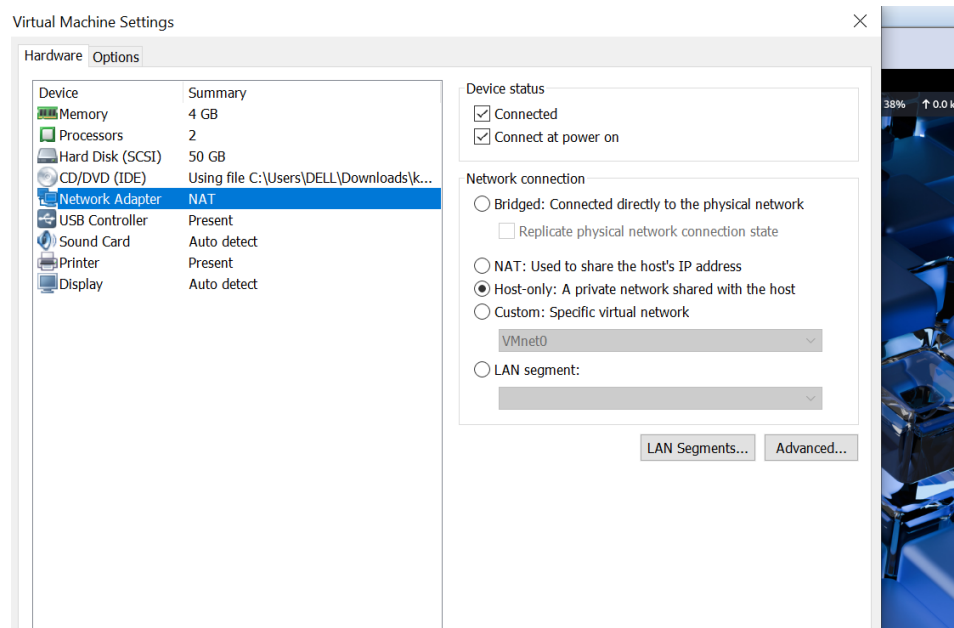
Objective The primary goal of this task is to establish a secure and isolated Purple Team laboratory environment. By utilizing virtualization technology, the environment is designed to contain both an attacker machine and a victim machine. This setup facilitates the simulation of cybersecurity scenarios within an internal network, ensuring that all activities remain contained and do not interfere with external production systems.

Tools and Environment

- Virtualization Platform: VMware Workstation
- Attacker Machine: Kali Linux
- Victim Machine: Windows 10

Laboratory Architecture The network topology is configured using a Host-Only architecture to ensure complete network isolation from the outside world.

- Network Connection: VMnet1
- IP Addressing Scheme:
 - Attacker Machine (Kali Linux): 192.168.72.128
 - Victim Machine (Windows 10): 192.168.72.129



```
usama@kali: ~  
(usama@kali)~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:38:65:4b brd ff:ff:ff:ff:ff:ff  
    inet 192.168.72.128/24 brd 192.168.72.255 scope global dynamic noprefixroute eth0  
        valid_lft 1387sec preferred_lft 1387sec  
    inet6 fe80::20c:29ff:fe38:654b/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

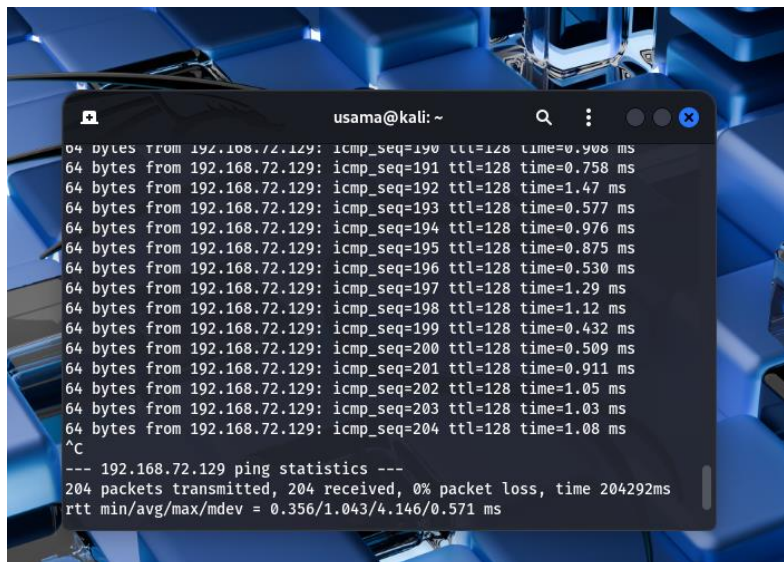
Implementation and Configuration

1. Virtual Machine Creation: Two distinct virtual machines were created within VMware Workstation. Hardware resources, including RAM, CPU, and Disk, were allocated appropriately for the specific operating systems (Kali Linux and Windows 10).
2. Network Setup: The network adapters for both machines were configured to use the Host-Only connection mode (VMnet1). Static IP addresses were manually assigned to each machine to ensure persistent and predictable communication paths.

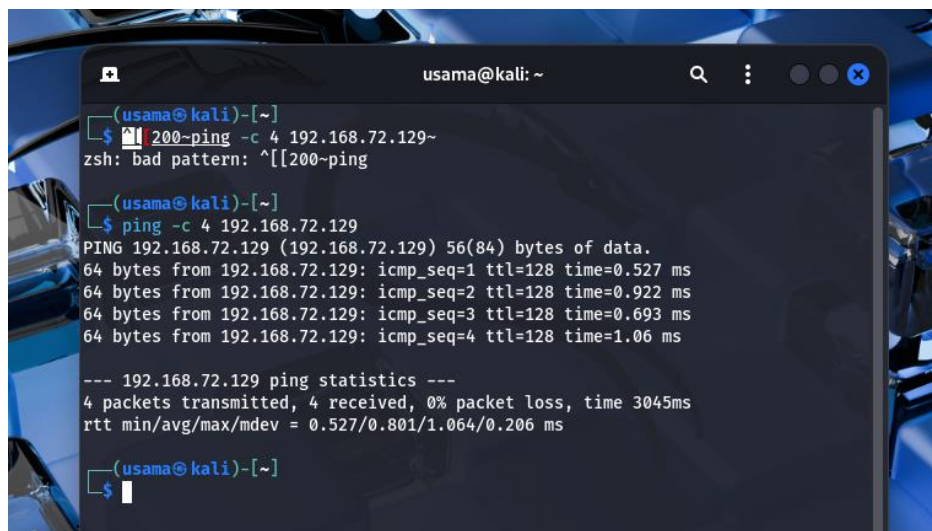
```
Recycle Bin Command Prompt  
Microsoft Windows [Version 10.0.19045.3803]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\uu497>ipconfig  
Windows IP Configuration  
  
Ethernet adapter Ethernet0:  
  
    Connection-specific DNS Suffix  . : localdomain  
    Link-local IPv6 Address . . . . . : fe80::4396:8bb:42f2:bdb1%6  
    IPv4 Address. . . . . : 192.168.72.129  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . :  
  
C:\Users\uu497>
```

Connectivity Verification To validate the network configuration, a connectivity test was performed. The ping command was executed from the attacker machine to the victim machine's IP address (192.168.72.129).

- **Result:** The test successfully demonstrated that the machines could communicate. Data packets were sent and received with 0% packet loss, confirming that the internal link is active and functioning correctly.



```
usama@kali: ~  
64 bytes from 192.168.72.129: icmp_seq=190 ttl=128 time=0.908 ms  
64 bytes from 192.168.72.129: icmp_seq=191 ttl=128 time=0.758 ms  
64 bytes from 192.168.72.129: icmp_seq=192 ttl=128 time=1.47 ms  
64 bytes from 192.168.72.129: icmp_seq=193 ttl=128 time=0.577 ms  
64 bytes from 192.168.72.129: icmp_seq=194 ttl=128 time=0.976 ms  
64 bytes from 192.168.72.129: icmp_seq=195 ttl=128 time=0.875 ms  
64 bytes from 192.168.72.129: icmp_seq=196 ttl=128 time=0.530 ms  
64 bytes from 192.168.72.129: icmp_seq=197 ttl=128 time=1.29 ms  
64 bytes from 192.168.72.129: icmp_seq=198 ttl=128 time=1.12 ms  
64 bytes from 192.168.72.129: icmp_seq=199 ttl=128 time=0.432 ms  
64 bytes from 192.168.72.129: icmp_seq=200 ttl=128 time=0.509 ms  
64 bytes from 192.168.72.129: icmp_seq=201 ttl=128 time=0.911 ms  
64 bytes from 192.168.72.129: icmp_seq=202 ttl=128 time=1.05 ms  
64 bytes from 192.168.72.129: icmp_seq=203 ttl=128 time=1.03 ms  
64 bytes from 192.168.72.129: icmp_seq=204 ttl=128 time=1.08 ms  
^C  
--- 192.168.72.129 ping statistics ---  
204 packets transmitted, 204 received, 0% packet loss, time 204292ms  
rtt min/avg/max/mdev = 0.356/1.043/4.146/0.571 ms
```



```
(usama@kali)-[~]  
$ !200~ping -c 4 192.168.72.129~  
zsh: bad pattern: ^[[200~ping  
  
(usama@kali)-[~]  
$ ping -c 4 192.168.72.129  
PING 192.168.72.129 (192.168.72.129) 56(84) bytes of data.  
64 bytes from 192.168.72.129: icmp_seq=1 ttl=128 time=0.527 ms  
64 bytes from 192.168.72.129: icmp_seq=2 ttl=128 time=0.922 ms  
64 bytes from 192.168.72.129: icmp_seq=3 ttl=128 time=0.693 ms  
64 bytes from 192.168.72.129: icmp_seq=4 ttl=128 time=1.06 ms  
  
--- 192.168.72.129 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3045ms  
rtt min/avg/max/mdev = 0.527/0.801/1.064/0.206 ms  
  
(usama@kali)-[~]  
$
```

Task 2: Packet Analysis and Protocol Deep Dive

Objective

The primary objective of this task was to analyze network traffic and understand the fundamental mechanics of data transmission across a network. This phase focused on capturing live packets, observing the communication flow between an attacker and a target system, and identifying the structural components of network protocols.

Tool Used: Wireshark

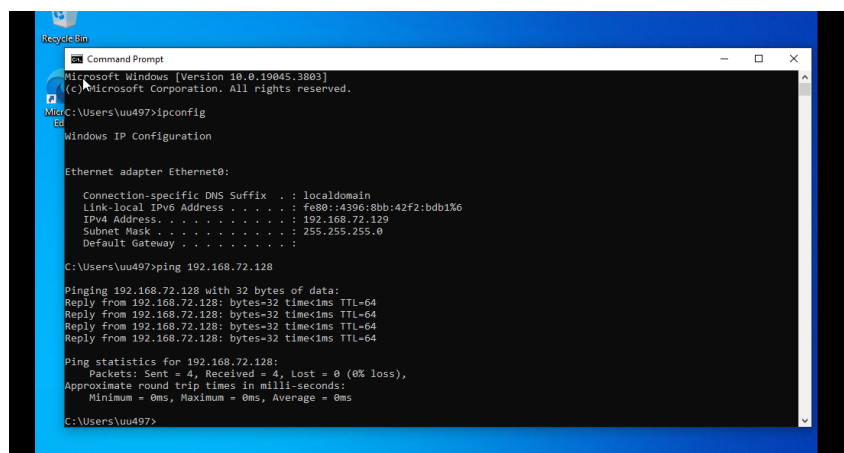
For this analysis, **Wireshark** was utilized as the primary network protocol analyzer. Wireshark provides deep visibility into the network layer, allowing for the real-time capture and inspection of individual data packets.

Packet Capture Methodology

The packet capture was performed on the **Kali Linux** machine using the following systematic approach:

- **Interface Selection:** The active network interface (**eth0**) was selected, which was configured on a **Host-Only** network to isolate the lab traffic.
- **Traffic Generation:** To create observable data, the **Ping** utility was used to send ICMP echo requests between the Kali Linux attacker machine and the Windows 10 target machine.
- **Protocol Filtering:** A display filter for **ICMP** was applied within **Wireshark** to isolate the relevant ping traffic from background network noise.

Results and Technical Analysis



```
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lup497>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

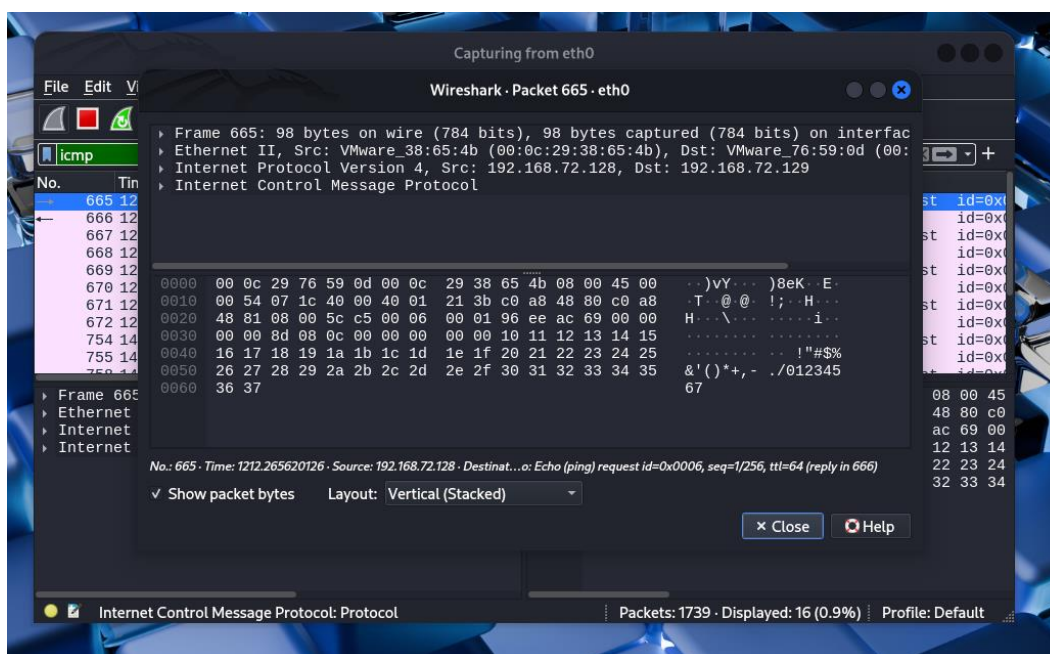
    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::4396:8bb:42f2:bdb1%6
    IPv4 Address. . . . . : 192.168.72.129
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

C:\Users\lup497>ping 192.168.72.128

Pinging 192.168.72.128 with 32 bytes of data:
Reply from 192.168.72.128: bytes=32 time<1ms TTL=64
Reply from 192.168.72.128: bytes=32 time<1ms TTL=64
Reply from 192.168.72.128: bytes=32 time<1ms TTL=64
Reply from 192.168.72.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.72.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\lup497>
```



Upon executing the connectivity test, Wireshark successfully captured the interaction between the two virtual machines:

1. **Traffic Source & Destination:** The capture shows packets originating from the Kali Linux IP (**192.168.72.128**) and reaching the Windows 10 target IP (**192.168.72.129**).
2. **ICMP Echo Process:** The "Info" column identifies the standard ICMP handshake: an **Echo (ping) request** followed immediately by an **Echo (ping) reply** from the target.

3. **Protocol Header Inspection:** By selecting a specific packet, the Internet Control Message Protocol headers were analyzed. The "Type 8" (Request) and "Type 0" (Reply) codes confirm a successful bi-directional communication channel.
4. **Data Payload Analysis:** The Hexadecimal view reveals the data payload (e.g., 10 11 12 13...), which serves as a placeholder to verify that the data sent by the source is exactly what is received by the destination.

Conclusion :

The Purple Team laboratory environment has been successfully established, providing a robust and safe sandbox for security simulations. This task successfully demonstrated the network visibility required for professional security operations by capturing and analyzing ICMP traffic between the machines. The implementation confirms that the systems can communicate effectively via the **Host-Only** network while remaining completely isolated from external threats.